Active-layer thickness prediction and regional differences across permafrost regions using a physics-residual machine-learning workflow

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Abstract

Active-layer thickness (ALT) observations are sparse and clustered. A Stefan model applies to unobserved regions but uses one soil coefficient, whereas machine learning (ML) and ML-based products lose accuracy away from their training sites. We compared a physics-residual workflow, in which gradient boosting learns the residual of a locally recalibrated Stefan model from climate, terrain and soil inputs, against the Stefan model, ML trained on the inputs alone (direct ML) and five products, on 19,172 site-year observations from Alaska, the Lena Delta and north-western Canada, each with the climate of its year. Under spatial-block validation it reduced the Stefan error by 3.6 cm in the Lena Delta and 3.7 cm in Canada. With a region withheld, direct ML exceeded the Stefan error by 17.8 cm in the Lena Delta, whereas the physics-residual model stayed within 2.0 cm and gained 3.9 cm from 25 local labels, with which it also outperformed direct ML. On the same site-years it matched the ML products in Alaska and Canada and surpassed them in the Lena Delta. The yearly 1 km maps carry an area of applicability and prediction intervals, and sparsely observed regions gain a Stefan-based map that improves with representative local labels.

Keywords: permafrost, active-layer thickness, Stefan model, physics-guided machine learning, spatial cross-validation, spatial transferability

Introduction

The active layer is the near-surface ground above permafrost that thaws each summer and refreezes each winter. Its thickness (ALT), the maximum depth of seasonal thaw, controls ground stability, near-surface hydrology and the amount of soil organic carbon exposed to decomposition[1–3]. ALT varies with the thermal conditions of each year, and this inter-annual variation is recorded by long-term programmes such as the Circumpolar Active Layer Monitoring (CALM) network[4,5]. The observations, however, are few and clustered: most come from a few campaigns and long-term grids, many sites are visited once, and large parts of the permafrost region have none[6,7]. Any ALT map therefore depends on a model that extends sparse observations to unobserved cells and years, and its reliability is set by the accuracy of that model away from them.

Two types of model are used for this extension. The Stefan solution relates the thaw depth to the thawing degree-days (TDD) of the season through one coefficient that summarises the thermal properties and moisture of the soil[4,8,9]. It responds to the climate of each year and, having one parameter, can be refitted with few observations and applied to distant regions, but one coefficient per region cannot represent the soils and terrain within the region, which leave systematic residuals. Machine learning (ML) has the opposite strengths and weaknesses: it fits the training sites closely from climate, terrain and soil predictors, but it must learn the response to climate from the training observations (labels), and its accuracy falls at locations unlike the training sites when it is validated with spatially separated data[10–13]. The gridded ALT products that now cover the northern permafrost region follow one of these two approaches. The European Space Agency Climate Change Initiative (ESA CCI) product solves a ground heat-transfer model driven by satellite land-surface temperature[14,15]; the ML products of Liu et al.[16] and Wei et al.[17] relate observations of single sites in single years (site-years) to yearly predictors, and Aalto et al.[18] and Ran et al.[19] map a multi-year mean. The ML products are validated by random or leave-one-site-out splits within their own site sets; because the validation data differ, their reported errors (21.6 cm[16] to 60.5 cm[17]) are not comparable, and none has been evaluated against observations of a stated year in a region without training sites.

Physics-guided learning, in which a physical model carries the dominant response and a learner corrects what the physics omits, has improved accuracy and extrapolation in lake, river and soil applications[20–22], and in Alaska the Stefan model retained its accuracy on held-out sites better than a random forest[23]. Whether a learner that corrects the Stefan model keeps the transferability of the physics (its accuracy in regions not used for training) while gaining the accuracy of ML has not been tested for ALT. Its practical value depends on three points that have not been established: the conditions under which the correction improves on the physics, the number of local observations a new region needs, and its accuracy relative to ML trained on the covariates alone (direct ML) and to the published products when all are evaluated on the same sites and years.

Here we address these three points with a site-year design, in which every label, climate input, prediction and score refers to one site in one stated year, with the climate input computed from ERA5-Land for that year and accumulated to the observation

day[24]. With 19,172 site-year observations of 2010–2024 from Alaska, the Lena Delta and north-western Canada, we compare the Stefan model, direct ML and a physics-residual workflow, in which gradient boosting learns the residual of a Stefan model recalibrated on local labels from 23 climate, terrain and soil inputs, under five validation designs from random splits to the hold-out of an entire region, and we evaluate five published products on the same site-years. The study (i) shows how and why the ranking of the methods changes with the validation design and under which conditions the accuracy of direct ML degrades, (ii) identifies the conditions and the number of local labels under which the residual learner improves on the physics, (iii) explains the regional differences between the published products, and (iv) provides yearly 1 km ALT maps of the three regions for 2011–2024 with an area of applicability (the covariate range supported by the training data) and prediction intervals.

Data and methods

Study regions and active-layer observations

We used three regions with dense, multi-year thaw-depth observations: Alaska, the Lena Delta and north-western Canada (Canada (NW) in the figures and tables; Fig. 1, Table 1). Observations for Alaska and Canada were taken from the ABoVE soil moisture and active-layer thickness compilation, version 2[6], and from the CALM end-of-season grid values[5]; observations for the Lena Delta were taken from the ALLena thaw-depth data set[7]. ABoVE probe measurements were restricted to August–October, because earlier measurements fall well short of the seasonal maximum, and to values greater than 0 and less than 300 cm. ALLena measurements were restricted to August–September values without the probe-limit flag (which marks measurements that reached the probe length); the seasonal maximum of each location and year was taken as the label and the mean day of its measurements as the observation day. Observations were grouped by site, defined as a location with coordinates rounded to 0.0001° (about 10 m), and averaged within each site and year, retaining the mean day of observation; probe points of the ABoVE transects are therefore separate sites, and several sites can lie in one 1 km cell. CALM thaw depths are probed near the end of the thaw season, in late August or early September[4,25], so CALM values without a date were assigned day 244 (1 September). We kept the years 2010–2024, for which year-specific climate inputs are available. The resulting labels comprise 15,540 site-years in Alaska (2010–2024), 2,436 in the Lena Delta (2010–2022) and 1,196 in Canada (2010–2019 and 2021–2024), distributed over 73, 16 and 26 blocks of 0.5° (Table 1). The sites lie in 321, 143 and 71 cells of 1 km (910, 244 and 144 cell-years); up to 701 sites of one transect fall in one cell and year, so that site-years are not independent samples at the 1 km scale, which the spatial-block design accounts for. The three programmes differ in protocol (probe transects in ABoVE and ALLena, 1 km grid means of about 100 probe points in CALM) and in site selection; all were used as reported. Throughout the study, every label, climate input, prediction and score refers to one site in one stated year: a label of year y is paired only with the climate of year y, and a map of year y uses only the climate of year y. Sources, versions and licences are listed in Supplementary Table S1, the input variables in Supplementary Table S2,

Table 1 Data and years used. Top: ALT labels by region (site-years of 2010–2024 with year-specific climate input; a site is an observation location with coordinates rounded to 0.0001°, and the 1 km cells are the map cells that contain the sites; blocks are the 0.5° cells used for spatial cross-validation and bootstrap; there are no Canadian labels in 2020). Bottom: inputs and products, with their spatial domain, resolution (ERA5-Land 0.1° is about 9 km; Aalto et al. 30″ about 1 km), the period available and the years used in this study.

Region	Site-years	Sites	1 km cells	Blocks	Years	Sites ≥4 yr	Largest year
Alaska	15,540	13,253	321	73	2010–2024	203	2014 (51%)
Lena Delta	2,436	2,206	143	16	2010–2022	25	2016 (28%)
Canada (NW)	1,196	730	71	26	2010–2024	48	2022 (37%)

Data	Domain	Resolution	Available	Used	Role
ERA5-Land monthly[24]	global	0.1°	1950–	2010–2024 (mean 2011–2024 in Fig. 3a)	Climate input
Copernicus DEM[26]	global	30 m	static	–	Terrain
SoilGrids 2.0[27]	global	250 m	static	–	Soil
MODIS MOD13Q1, MOD10A2[28,29]	global	250, 500 m	2000–	2010–2024	Sensitivity
ESA CCI permafrost probability[14]	N. Hemisphere	1 km	1997–2021	mean 1997–2021	Map domain (≥ 50%)
Wei et al.[17]	N. Hemisphere	1 km	yearly 2000–2024	2010–2024	Comparison
Liu et al.[16]	N. Hemisphere	1 km	yearly 2003–2020	2010–2020	Comparison
ESA CCI ALT v4[14]	N. Hemisphere	1 km	yearly 1997–2021	2010–2021	Comparison
ESA CCI ALT v5[15]	N. Hemisphere	1 km	yearly 1997–2023	2010–2023	Comparison
Aalto et al.[18]	circum-Arctic	30″	mean 2000–2014	one value per cell	Comparison
Yi and Kimball[30]	Alaska	1 km	yearly 2001–2015	2010–2015	Reference, not scored

the years used in each analysis in Supplementary Table S3 and the processing steps in Supplementary Methods S1.

Year-specific climate input

Monthly ERA5-Land fields for January 2010 to December 2024[24] (2 m air temperature, top-layer soil temperature and snow water equivalent) were extracted at the nearest land grid cell of each site. For site c and year y, the thawing degree-day sum[4,9] is

$$\mathrm{TDD}_{c,y} = \sum_{m=1}^{12} \max(T_{c,y,m},\, 0)\, d_{y,m}, \tag{1}$$

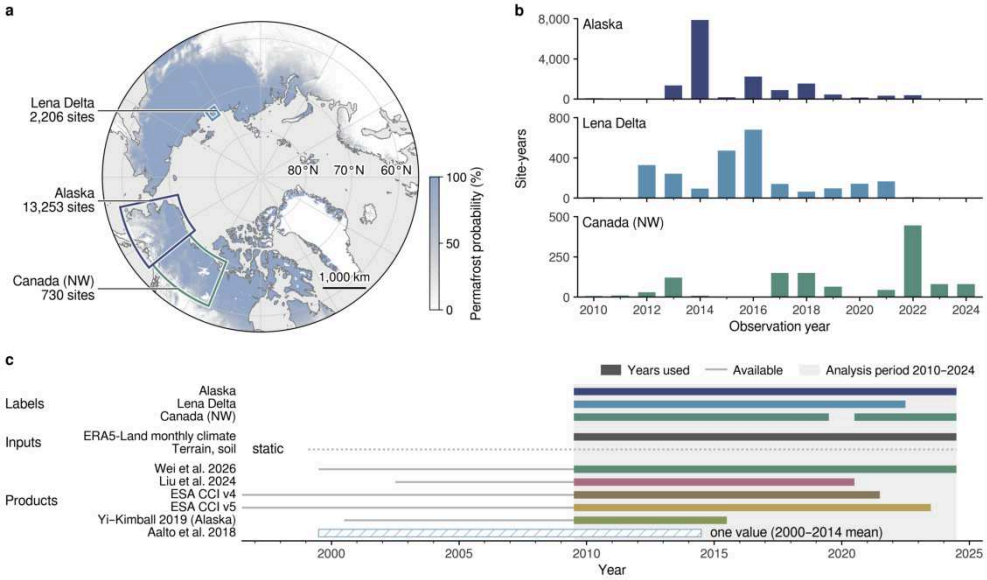


Fig. 1 Study regions, observations and data periods. a, ESA CCI permafrost probability (mean of 1997–2021) with the three study regions and the number of sites with ALT labels in 2010–2024. **b**, Site-year labels per observation year. **c**, Periods covered by the labels, the ERA5-Land climate input, the static covariates and the six published products; dark bars mark the years with product values at the label sites, and the shaded band marks the analysis period.

where $T_{c,y,m}$ is the monthly mean air temperature and $d_{y,m}$ the number of days in month m. Because most observations were made before the end of the thaw season, the model input is the TDD accumulated to the day of observation t,

$$\mathrm{TDD}^{\mathrm{obs}}_{c,y} = C_{c,y,k-1} + \frac{t - t_{k-1}}{d_{y,k}} \left(C_{c,y,k} - C_{c,y,k-1}\right), \tag{2}$$

where $C_{c,y,k}$ is the cumulative TDD to the end of month k, k is the month containing t and t_{k-1} is the last day of the preceding month. Seven further climate variables were computed for each site and year: the mean annual air temperature, $\sqrt{\mathrm{TDD}}$, the warmest and the coldest monthly mean temperature, the freezing degree-days of the preceding winter (October of the previous year to April of the observation year), the mean top-layer soil temperature and the mean snow water equivalent of the hydrological year ending in September of the observation year (Supplementary Table S2, Supplementary Methods S2); because the series starts in January 2010, the winter variables of 2010 were computed from January–April (freezing degree-days) and January–September (snow water equivalent) of 2010. Two alternative inputs were evaluated on the same site-years: the full-season TDD of the observation year, and the 2011–2024 mean of the yearly values (mean climate input), which corresponds to the multi-year inputs of existing products. Because most observations are made in August, the TDD accumulated to the observation day at the label sites is a median fraction of 0.73, 0.81 and 0.89 of the full-season sum in Alaska, the Lena Delta and

Canada. Yearly maps use the TDD accumulated to day of year 244 (1 September in non-leap years), the end-of-season reference date of the CALM protocol; this date is 10 days later than the upper quartile of the pooled observation days (224–234) and within their full range (213–288). Map values thus refer to the thaw depth reached by 1 September of the stated year.

Static covariates

Six terrain variables (elevation, slope, the sine and cosine of aspect, topographic position index and roughness) were computed from the 30 m Copernicus DEM[26] in a 33×33 pixel window (about 1 km) around each site. Nine soil variables (clay, sand, silt, bulk density, coarse fragments and pH at 5–15 cm, and soil organic carbon at 0–5, 5–15 and 15–30 cm) were sampled from SoilGrids 2.0 at 250 m[27]. The main feature vector $\mathbf{x}$ therefore contains 23 variables (eight climate, six terrain and nine soil). Sensitivity tests used five alternative covariate sets: climate only; climate and terrain; the main set plus MODIS summer NDVI, EVI and snow-off timing for 2010–2024 (MOD13Q1 and MOD10A2, version 6.1)[28,29]; the main set plus high-resolution soil carbon and land-cover classes[31]; and all variables including the climate of the preceding year. The ESA CCI permafrost probability (mean of 1997–2021)[14] was used only to define the map domain (probability of at least 50%) and not as a predictor.

Model formulation

The Stefan model describes the thaw depth as

$$\widehat{\mathrm{ALT}}_{\mathrm{S}} = E\,\sqrt{\mathrm{TDD}^{\mathrm{obs}}}, \tag{3}$$

where E ($\mathrm{cm}\,(^{\circ}\mathrm{C\,d})^{-1/2}$) summarises the thermal properties and moisture of the soil[8,9]. The relation holds within a season at the Lena Delta monitoring sites and, with more scatter, among years at cells observed repeatedly (Supplementary Fig. S1). We chose the Stefan model as the physical anchor because it has one parameter, which can be refitted from few observations and transferred between regions, and because its input is the year-specific TDD; the Kudryavtsev model[32,33] with a scaling coefficient and a non-negative combination of both models were evaluated as alternative anchors (Supplementary Table S4). Fitting the coefficient, often called the edaphic factor, to local observations follows the practice of Stefan-based ALT mapping[4,34]. For a set of n training labels y_i with $s_i = \sqrt{\mathrm{TDD}_i^{\mathrm{obs}}}$, the coefficient is recalibrated by least squares through the origin,

$$\hat{E} = \frac{\sum_{i=1}^{n} y_i\, s_i}{\sum_{i=1}^{n} s_i^2}. \tag{4}$$

The source-coefficient Stefan model uses $\hat{E}$ fitted on the other two regions, and the recalibrated Stefan model uses $\hat{E}$ fitted on the training labels of the target region. The physics-residual model retains the recalibrated Stefan prediction as an anchor

and adds a learned correction, the residual form of physics-guided learning[20] (Fig. 2):

$$\widehat{\mathrm{ALT}} = \hat{E}\,s + \lambda\, g(\mathbf{x}), \qquad g = \arg\min_{f} \sum_{i=1}^{n} \left[\left(y_i - \hat{E}\,s_i \right) - f(\mathbf{x}_i) \right]^2, \tag{5}$$

where g is a CatBoost gradient-boosting regressor[35] (600 iterations, learning rate 0.03, depth 6, L2 regularisation 3) and $\lambda = 0.5$ shrinks the correction towards the physical anchor. CatBoost was used as the default learner because gradient-boosted trees need no feature scaling and are the usual reference for tabular data of this size[36]; random forest, a multilayer perceptron, TabM and FT-Transformer[37–39] were run in its place as a sensitivity test, and no learner was best in all regions (Supplementary Fig. S2). The value $\lambda = 0.5$ was chosen as a compromise between the within-region gain (the reduction in spatial-block RMSE relative to the recalibrated Stefan model), which increased with λ (Lena Delta −5.1 cm at $\lambda = 1$), and the error under region hold-out (defined below), which also increased with λ (Lena Delta +5.4 cm at $\lambda = 1$); $\lambda = 0.25$ and 1 are reported as sensitivity cases. Three alternative structures were compared: direct ML, which uses no physical information, and two structures that place it elsewhere,

$$\begin{aligned} &\widehat{\mathrm{ALT}}_{\mathrm{D}} = f_{\mathrm{D}}(\mathbf{x}), \qquad \widehat{\mathrm{ALT}}_{\mathrm{I}} = f_{\mathrm{I}}\left(\mathbf{x},\, \hat{E}\,s,\, \widehat{\mathrm{ALT}}_{\mathrm{K}}\right), \\ &\widehat{\mathrm{ALT}}_{\mathrm{A}} = f_{\mathrm{A}}(\mathbf{x}) \text{ fitted to } \{(\mathbf{x}_i, y_i)\}_{i=1}^{n} \cup \{(\mathbf{x}_j, \hat{E}\,s_j)\}_{j=1}^{n}, \end{aligned} \tag{6}$$

where direct ML (f_{D}) fits CatBoost to ALT from $\mathbf{x}$ alone, physics-input ML (f_{I}) adds the recalibrated Stefan prediction and the Kudryavtsev[32] prediction $\widehat{\mathrm{ALT}}_{\mathrm{K}}$ as features, and the physics pseudo-label model (f_{A}) adds to the n training samples an equal number of samples j drawn from cells of the evaluation fold, each with its covariates as input and the Stefan prediction fitted on the training fold as the target. All learned models were fitted with three random seeds and their predictions averaged (the transfer models without target labels with one seed). SHAP values (SHapley Additive exPlanations; the contribution of each input to a prediction)[40] were computed for the physics-residual model and direct ML fitted to all labels of each region.

Validation designs and uncertainty

All models were evaluated on out-of-fold predictions, that is, predictions of each site-year by a model fitted without the fold that contains it, under five designs[10,11]: random five-fold cross-validation by site-year; spatial-block five-fold cross-validation, with 0.5° blocks assigned to folds balanced by the number of site-years (main design); year hold-out, in which each observation year with at least 20 site-years is predicted by models trained on the other years; block-by-year hold-out, in which both the block fold and the year of the scored site-years are excluded from training (folds with at least 10 site-years); and region hold-out, in which a region is predicted by models trained on the other two regions after removing training cells within 100 km of the test region.

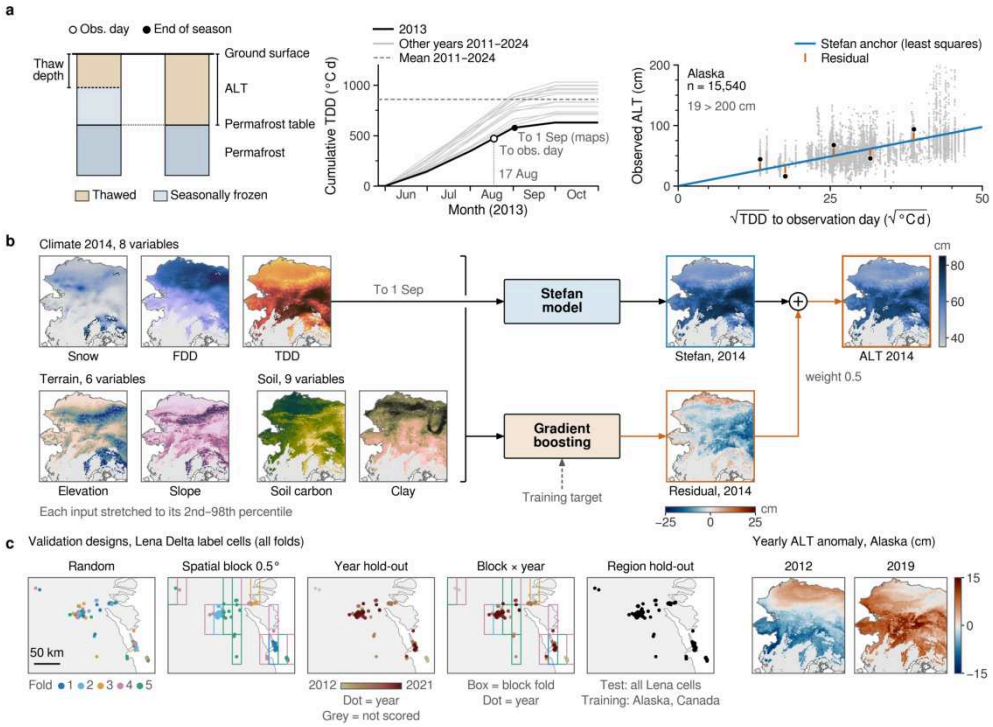


Fig. 2 Site-year design, Stefan anchor, physics-residual model and validation designs. **a**, Thaw depth on the observation day and ALT at the end of the season (left); cumulative thawing degree-days of a Lena Delta cell in 2013, with the model input accumulated to the observation day and the map input to 1 September (centre); observed ALT against $\sqrt{\mathrm{TDD}_{\mathrm{obs}}}$ for the Alaska site-years with the Stefan anchor and five residuals (right). **b**, Physics-residual model: the Stefan anchor plus gradient boosting of its residual from 23 climate, terrain and soil inputs (equation (5)), illustrated with the Alaska map of 2014. **c**, Validation designs on the Lena Delta label cells (colour marks the fold, the observation year or the test region), and ALT anomalies of the Alaska map in 2012 and 2019.

Accuracy is expressed as

$$\mathrm{RMSE} = \left[\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2\right]^{1/2}, \qquad \Delta\mathrm{RMSE}_{A,B} = \mathrm{RMSE}_A - \mathrm{RMSE}_B, \tag{7}$$

where both terms of ΔRMSE are computed on the same site-years; negative values favour method A. Uncertainty was quantified with a paired block bootstrap[10,41]: the 0.5° blocks were resampled with replacement $B = 2000$ times (1000 times for the label-number experiments and for part of the covariate-set and learner sensitivity runs), ΔRMSE was recomputed from all site-years of the resampled blocks, and the 2.5th and 97.5th percentiles of the replicates define the 95% interval,

$$\mathrm{CI}_{95} = \left[Q_{0.025}\left(\Delta\mathrm{RMSE}^{(b)}\right),\ Q_{0.975}\left(\Delta\mathrm{RMSE}^{(b)}\right)\right], \quad b = 1, \ldots, B. \tag{8}$$

We describe a difference as resolved when its interval excludes zero. Because several sites can lie in one 1 km cell (median 5, 3 and 5 sites per cell in Alaska, the Lena Delta and Canada), the main comparisons were also scored on cell-year means of the observations and predictions (Supplementary Table S5). Conditions of the gain were examined within regional terciles of eight covariates (full-season TDD of the observation year, freezing degree-days, snow water equivalent, soil organic carbon, clay, slope, elevation and summer NDVI); terciles with fewer than 30 site-years or fewer than five blocks were not evaluated. Year-wise RMSE was computed for years with at least 30 site-years (Supplementary Fig. S7).

Label-number and transfer experiments

Within each region, the training labels of each spatial-block fold were subsampled to n = 25, 50, 100, 250, 500, 1000, 2500 or all (three random draws), the Stefan coefficient was recalibrated on the same n labels, and the complete test fold was scored. For transfer, each region was scored in its spatial-block folds by models trained on all labels of the other two regions plus $n = 0$, 25, 50, 100, 250 or 500 labels drawn from the training folds of the target region; the Stefan coefficient was refitted on the n target labels, and the residual learner was trained on the source-region samples anchored with the source coefficient together with the target-region samples anchored with the refitted coefficient,

$$\begin{aligned} \widehat{\mathrm{ALT}}_{\mathrm{T}} &= \hat{E}_{\mathrm{T}}\, s + \lambda\, g_{\mathrm{ST}}(\mathbf{x}), \\ g_{\mathrm{ST}} &= \arg\min_{f} \Big[\sum_{i \in \mathrm{S}} \big(y_i - \hat{E}_{\mathrm{S}}\, s_i - f(\mathbf{x}_i)\big)^2 + \sum_{i \in \mathrm{T}} \big(y_i - \hat{E}_{\mathrm{T}}\, s_i - f(\mathbf{x}_i)\big)^2 \Big], \end{aligned} \tag{9}$$

where S denotes the source-region samples, T the n target-region samples, $\hat{E}_{\mathrm{S}}$ the coefficient fitted on the source samples and $\hat{E}_{\mathrm{T}}$ the coefficient refitted on the target samples ($\hat{E}_{\mathrm{T}} = \hat{E}_{\mathrm{S}}$ for $n < 3$); in this experiment, source-region samples within 100 km of the target region were not removed.

Comparison with published products

Six publicly archived products were downloaded and sampled at the label sites (nearest 1 km pixel; Supplementary Methods S3 and Table S1): the Northern Hemisphere products of Wei et al.[17] (1 km, yearly 2000–2024), Liu et al.[16] (1 km, yearly 2003–2020), ESA CCI Permafrost ALT v4 (1997–2021) and v5 (1997–2023)[14,15] and Aalto et al.[18] (one value per cell for the 2000–2014 baseline), and the Alaska-only product of Yi and Kimball[30] (2001–2015), which was used as a reference and not scored because it covers one region and ends in 2015. For yearly products the value of the observation year was used, so that products and models are compared for the same site and the same year. Products and models were compared on common site-years, at which Wei et al., Liu et al. and ESA CCI v4 all provide a value for the observation year (2010–2020); ESA CCI v5 and Aalto et al. were scored on the common site-years where they have a value, and the year-wise comparison used all site-years of a year with a value of Wei et al. Because Wei et al. published the coordinates of their training

sites, comparisons were repeated after excluding label cells within 25 km of these sites (leakage mask). Liu et al. trained their random forest on CALM and on the ABoVE soil-moisture and active-layer compilation[16], an earlier release of the Alaska and Canada observations used here, so their values at these sites are not independent of our labels; the training sites of Wei et al. overlap with our Alaska sites in the same way. To test whether the residual learner can also correct a product, the value p of the Wei et al. product for the observation year was recalibrated on the training fold,

$$\hat{p} = a + b\,p, \tag{10}$$

and the residual $y - \hat{p}$ was learned with the same covariates, learner, λ and folds as in equation (5) (Supplementary Methods S4).

Yearly mapping, area of applicability and prediction intervals

For each region, the recalibrated Stefan and physics-residual models were fitted to all site-years of the region. They were applied to the 1 km cells with an ESA CCI permafrost probability of at least 50% (the continuous and discontinuous permafrost zones of Obu et al.[42], where an active layer above permafrost exists and where the labels lie), excluding cells that are water (mean Global Surface Water occurrence of at least 50%) or lack an input (858,517, 38,086 and 1,151,169 cells remain). Each map year from 2011 to 2024 uses the climate of that year with the TDD accumulated to 1 September. The area of applicability followed Meyer and Pebesma[43]. With $\mathbf{z}$ the standardised 23-dimensional feature vector (with the 2011–2024 mean of the yearly map inputs), the dissimilarity of map cell k is $d_k = \min_j \|\mathbf{z}_k - \mathbf{z}_j\|_2$ over all training samples j, and the cell is inside the area of applicability when

$$d_k \le \tau, \qquad \tau = Q_{0.95}\left\{ \min_{j:\, b(j)\neq b(i)} \|\mathbf{z}_i - \mathbf{z}_j\|_2 \;:\; i \in \text{training samples} \right\}, \tag{11}$$

where $b(\cdot)$ denotes the spatial block. For year y and cell k, the anomaly is $a_{k,y} = \widehat{\mathrm{ALT}}_{k,y} - \overline{\widehat{\mathrm{ALT}}_k}$, where the overbar denotes the 2011–2024 mean; the regional mean anomaly is the mean of $a_{k,y}$ over the mapped cells, and the trend of each cell is the least-squares slope

$$\beta_k = \frac{\sum_y (y - \bar{y})\, a_{k,y}}{\sum_y (y - \bar{y})^2}, \tag{12}$$

expressed in cm per decade, as in previous yearly ALT products[16,30]. The 90% prediction interval of a map value is

$$\left[\widehat{\mathrm{ALT}}_{k,y} + Q_{0.05}(r),\ \widehat{\mathrm{ALT}}_{k,y} + Q_{0.95}(r) \right], \tag{13}$$

where r denotes the spatial-block out-of-fold residuals of the region; this is a split-conformal interval[44] of constant width within a region, and the dissimilarity d_k of each cell (equation (11)) indicates the uncertainty of individual cells. For display, the 1 km values were interpolated bilinearly onto a 250 m (Lena Delta) or 500 m

(Alaska, Canada) hillshade grid. For each region, the map of the year with the most observed 1 km cells having a value in our map and in both products (2013 in Alaska and Canada, 2016 in the Lena Delta) was compared with the two ML products on those cells, using the product values of the same year at the label sites, the cell means of the observations and our spatial-block out-of-fold predictions; the product maps are displayed on the cells of our map, and each region uses one colour range set by the 5th–95th percentiles of our map inside the area of applicability, extended to the product medians, with product values beyond it saturated.

Computation

Models were fitted with fixed random seeds on 96-core CPU nodes, and the neural learners on one NVIDIA A100 GPU; data preparation, aggregation and figures were run on a local server (Supplementary Methods S5).

Code availability

The code used for data processing, model fitting and evaluation is available at [repository and DOI]. Every number in the text is generated automatically from the stored result tables.

Results

Climate input, validation design and the in-region versus out-of-region contrast

Replacing the 2011–2024 mean climate input by the input of the observation year reduced the spatial-block RMSE of the recalibrated Stefan model in the Lena Delta (ΔRMSE −1.4 [−2.6, −0.8] cm) and that of the physics-residual model in the Lena Delta (−1.6 [−2.4, −1.0] cm) and Canada (−1.5 [−2.0, −0.4] cm), whereas in Alaska, where half of the labels come from one year, the change was not resolved (Fig. 3a). Under spatial-block cross-validation with this input, the physics-residual model reduced the RMSE of the recalibrated Stefan model from 21.0 to 17.4 cm in the Lena Delta (ΔRMSE −3.6 [−7.8, −0.8] cm) and from 30.6 to 26.9 cm in Canada (−3.7 [−4.8, −0.8] cm), whereas in Alaska their RMSEs were 18.6 and 18.4 cm (−0.2 [−0.6, +0.1] cm; Fig. 3b). The predictions of the recalibrated Stefan model span a narrow range in every region, because one coefficient per region distinguishes sites only through their TDD, whereas the predictions of the physics-residual model extend along the 1:1 line in the Lena Delta and Canada (Supplementary Fig. S3). Direct ML, which learns the whole response from the labels, had the lowest site-year RMSE when all labels of a region were used (Alaska 17.3 versus 18.4, Lena Delta 15.9 versus 17.4 and Canada 26.1 versus 26.9 cm for the physics-residual model). When the same predictions were scored on 1 km cell-year means, the unit of the maps, this advantage disappeared in Alaska (24.7 versus 24.0 cm) and narrowed elsewhere (Supplementary Table S5), which suggests that part of it comes from fitting differences between probe points inside a cell. The difference between direct ML and the recalibrated Stefan model was resolved only in Canada (−4.6 [−6.2, −0.2] cm).

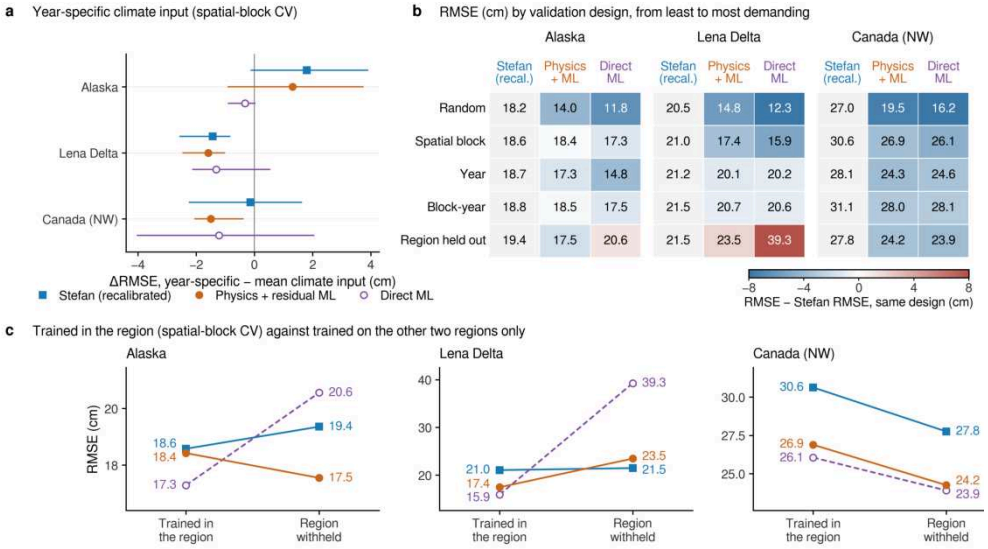


Fig. 3 Climate input, validation design and the in-region versus out-of-region contrast. **a**, Change in spatial-block RMSE when the 2011–2024 mean climate input is replaced by the input of the observation year (95% paired block-bootstrap intervals; negative values favour the year-specific input). **b**, RMSE of three methods under five validation designs, ordered from the least to the most demanding; the colour of a cell is its difference from the recalibrated Stefan model of the same design (blue, lower; red, higher), and in the region hold-out the Stefan model uses the coefficient of the other two regions. **c**, RMSE of the same methods when a region's own labels are used (spatial-block cross-validation) and when the region is withheld and the models are trained on the other two regions only.

The ranking of the methods depended on the validation design (Fig. 3b, Supplementary Table S4). In random cross-validation, in which test sites lie next to training sites, direct ML had its lowest error (11.8 cm in Alaska); spatial-block cross-validation raised its error by 5.4, 3.6 and 9.9 cm in Alaska, the Lena Delta and Canada, and the year and block-by-year designs raised it further. When an entire region was withheld, the direct-ML error in the Lena Delta rose to 39.3 cm, against 21.5 cm for the Stefan model with the coefficient of the other regions (ΔRMSE +17.8 [+8.0, +23.9] cm), whereas the physics-residual model stayed within 2.0 cm of the Stefan model there and improved on it in Canada (−3.5 [−5.1, −0.1] cm; Fig. 3c). As test sites move away from training sites, the errors of the three methods change differently: the one-parameter Stefan model changes little with the design; direct ML is the most accurate where test and training sites interleave and the least accurate under region hold-out in the Lena Delta; and the physics-residual model keeps most of the in-region gain while its region hold-out error stays close to that of its anchor. Because maps are needed where there are no observations, spatial-block cross-validation is the main design below and the region hold-out the test of transferability.

The Kudryavtsev model, tested as an alternative anchor, was less accurate than the recalibrated Stefan model under spatial-block validation in Alaska (21.7 versus 18.6 cm) and the Lena Delta (20.4 versus 21.0 cm), more accurate in Canada (25.4

versus 30.6 cm), and less accurate under the region hold-out (25.1, 21.8 and 35.4 cm against 19.4, 21.5 and 27.8 cm); the Stefan model was therefore kept (Supplementary Table S4).

Conditions of the physics-residual gain

The gain of the residual learner depended on the available covariates (Fig. 4a). With climate covariates only, the residual model did not improve on the Stefan model in Alaska (+5.9 [−0.3, +14.2] cm) or Canada (+3.9 [−5.1, +10.3] cm); adding terrain and soil covariates produced the gains reported above in the Lena Delta and Canada, but not in Alaska. MODIS vegetation and snow timing or high-resolution soil and land-cover classes changed the result by less than 0.5 cm in the Lena Delta and Canada, and the latter added a small gain in Alaska (−0.8 [−1.4, −0.2] cm). Across regional terciles of eight covariates, the gain occurred in most conditions of the Lena Delta (19 of 23 terciles with intervals excluding zero) and Canada (17 of 23) and increased with the error of the Stefan model (correlation across terciles −0.99 in the Lena Delta and −0.79 in Canada; Fig. 4b), reaching −6.2 [−12.0, −0.7] cm in the warmest TDD tercile of the Lena Delta. In Alaska no tercile changed by more than 0.9 cm, so the absence of a gain was not confined to particular conditions (see Discussion). Terrain variables accounted for 50% of the mean absolute SHAP value of the residual model in the Lena Delta and soil variables for 39% in Canada (Supplementary Fig. S4). Of the structures that place the physics elsewhere, the pseudo-label model was close to the Stefan model (Lena Delta −0.3 [−0.4, −0.1] cm) and the physics-input model was close to direct ML. Replacing CatBoost by four other learners did not change the sign of the gain in the Lena Delta and Canada (Supplementary Fig. S2).

Number of labels within a region and transfer to a new region

The number of local observations that the learner needs was tested within regions and in transfer. With 25 training labels drawn from the site-years of a region, the physics-residual model was at least as accurate as a Stefan model recalibrated on the same labels (Alaska −0.9 [−1.9, +0.1], Lena Delta −0.8 [−2.2, −0.1] and Canada −0.3 [−3.2, +1.7] cm), and with 100 labels its gain was resolved in all three regions (−0.7 [−1.4, 0.0], −2.0 [−5.3, −0.3] and −1.9 [−3.7, −0.3] cm; Supplementary Fig. S5). Direct ML trained on 25 labels had a higher site-year RMSE than the Stefan model in Alaska (+2.8 [−0.3, +4.5] cm) and Canada (+1.2 [−1.5, +3.4] cm) and overtook the physics-residual model only with several hundred labels in Alaska and the Lena Delta. With thousands of labels direct ML had the lower error, whereas the physics-residual model reached almost the same accuracy with a fraction of the labels and was at least as accurate as the recalibrated Stefan model at every label number.

In the transfer experiment, each region was predicted by models trained on the other two regions and n labels of the target region (Fig. 4c; $n \leq 500$, drawn from the training folds of the target region). Without target labels, direct ML reached 39.5 cm in the Lena Delta, against 21.5 cm for the source-coefficient Stefan model and 23.8 cm for the physics-residual model; 25 target labels reduced the physics-residual error to 19.9 cm and the direct-ML error to 21.3 cm. In Alaska the physics-residual model

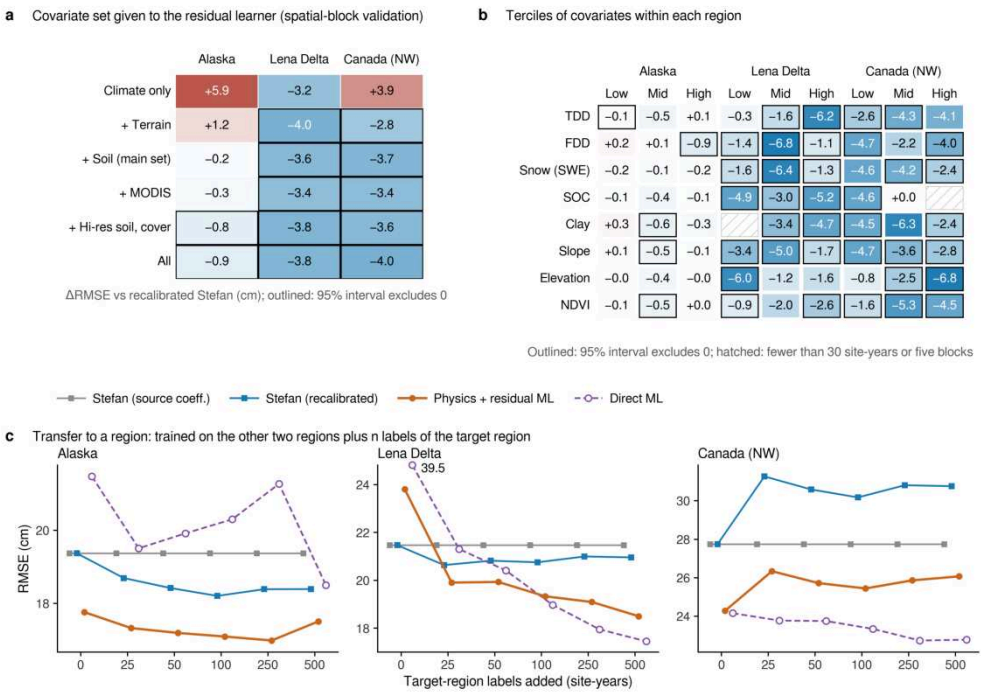


Fig. 4 Conditions of the physics-residual gain and transfer to a region with few labels. **a**, Difference in spatial-block RMSE between the physics-residual model and the recalibrated Stefan model for six covariate sets (value and colour of each cell; outlined cells have intervals excluding zero). **b**, The same difference within regional terciles of eight covariates; outlined cells have intervals excluding zero, hatched cells fewer than 30 site-years or five blocks. **c**, RMSE in each region for models trained on the other two regions plus n labels of the target region; a value beyond the axis is printed. Intervals in a and b are 95% paired block-bootstrap intervals; negative differences favour the physics-residual model.

gave the lowest error at every n (17.8 cm without target labels). In Canada, direct ML trained on the source regions was the most accurate (24.2 cm). Recalibrating the Stefan coefficient on 25 Canadian labels increased its error from 27.7 to 31.3 cm, and the error of the physics-residual model, which uses this anchor, increased from 24.3 to 26.3 cm. We attribute this to the difference in thaw depth between the two Canadian data sources (mean 63.4 cm at the probe transects and 89.1 cm at the CALM grids), which one coefficient fitted to 25 labels cannot represent.

Comparison with published products on the same sites and years

The published products were evaluated on the same site-years as our models. On the 461 common site-years of the Lena Delta (2010–2020; site-years at which Wei et al., Liu et al. and ESA CCI v4 all provide a value for the observation year), the physics-residual model reached an RMSE of 21.1 cm, against 28.4 cm for ESA CCI v5, 30.6 cm for Aalto et al., 31.0 cm for ESA CCI v4, 33.0 cm for Liu et al. and 34.6 cm for

Wei et al. (Fig. 5a, Supplementary Table S6). More than 25 km from the training sites of Wei et al., the paired differences in the Lena Delta were −12.5 [−15.6, −5.8] cm relative to Wei et al. and −5.0 [−9.5, −2.5] cm relative to ESA CCI v5 (802 and 2,352 site-years). In Alaska (17.1 versus 17.9 and 18.8 cm for Wei et al. and Liu et al.) and Canada (25.8 versus 27.0 and 27.9 cm) the model and the two ML products had similar errors.

In the Lena Delta the two ML products predicted between about 40 and 70 cm at most sites regardless of the observed ALT (Fig. 5b): the correlation between observed and predicted values was 0.05 for Wei et al. and 0.07 for Liu et al., against 0.73 for the physics-residual model, and their mean bias was +21.9 and +19.8 cm, against +2.8 cm, with the largest overestimation at the thin active layers of the northern delta (+21.1 and +19.9 cm north of 72°N). The error distributions (Fig. 5c) separate this systematic bias from the spread: in the Lena Delta the errors of the two ML products are shifted by about 20 cm with a spread similar to that of our model, whereas in Alaska and Canada all series are centred near zero and differ mainly in spread. The regional contrast corresponds to the location of the training sites of the products. Alaska label sites lie at a median distance of 7 km from the training sites of Wei et al., and Liu et al. trained their model on an earlier release of the Alaska and Canada observations used here, so their values at these sites are not independent of our labels; the Lena Delta sites lie 107 km (median) from the training sites of Wei et al. (Supplementary Fig. S6). A model trained without any Lena Delta label was already more accurate there than Wei et al. on the same site-years (26.0 versus 30.3 cm; ΔRMSE −4.3 [−6.1, −0.8] cm), so that the advantage in this region does not depend on local labels alone. Year by year, the physics-residual model had a lower error than Wei et al. in all 6 Lena Delta years with at least 30 site-years and in 7 of 13 Alaska years (Supplementary Fig. S7).

Maps of the year with the most observed cells in each region (2013 in Alaska and Canada, 2016 in the Lena Delta) show these differences (Fig. 6). At the 126 Alaska cells, the RMSE of the three maps was similar (14.9, 13.4 and 12.9 cm). The correlation with the observations was lower for our model (0.34) than for the products (0.53 and 0.57). Of these cells, 59 lie within 25 km of a training site of Wei et al.; at the 67 cells beyond that distance the RMSE of the three maps was 9.1, 10.9 and 8.5 cm and the correlation 0.30, 0.27 and 0.44. Above 600 m between 68 and 70°N, the products are deeper than our map by +18.7 and +52.9 cm. In the Lena Delta (14 cells observed in 2016) both products were deeper than the observations (mean bias +23.6 and +27.3 cm; RMSE 26.2 and 30.4 cm against 12.7 cm). In Canada (NW), the RMSE at the 33 observed cells was 23.7 cm for our model, 29.9 cm for Wei et al. (mean bias +20.5 cm) and 20.2 cm for Liu et al., whose training data include the earlier release of these observations; at the 27 cells beyond 25 km from a training site of Wei et al. the values were 17.5, 31.4 and 21.0 cm.

The residual learner can also be anchored to a product (equation (10)): in the Lena Delta the Stefan-anchored model was more accurate than the model anchored to the Wei et al. product (−2.8 [−6.2, −0.1] cm; Supplementary Fig. S8), and in Alaska and Canada the two anchors were not distinguishable.

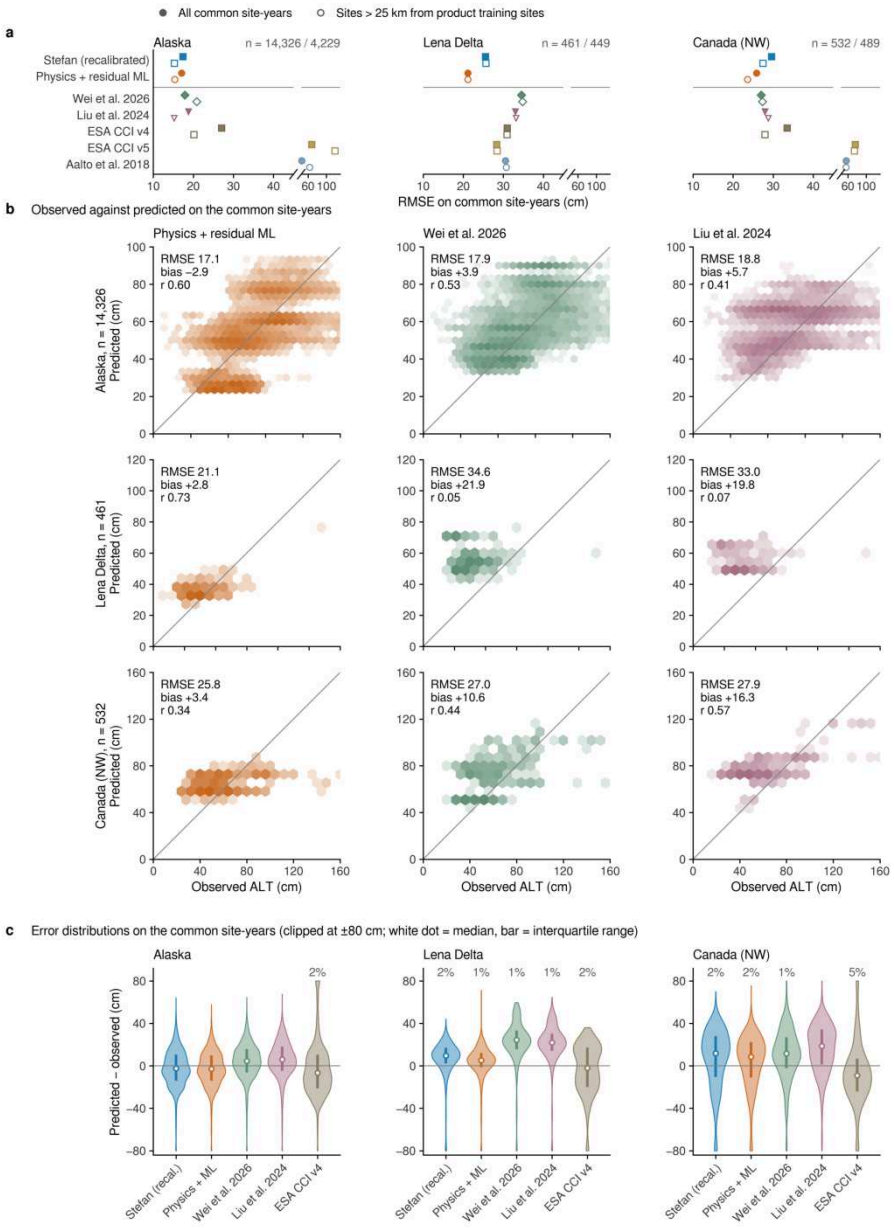


Fig. 5 Comparison with published products on the same sites and years. a, RMSE of the recalibrated Stefan model, the physics-residual model and five products on the common site-years (2010–2020), for all sites (filled) and for sites more than 25 km from the training sites of Wei et al. (open); values above 45 cm are drawn on the compressed segment. **b**, Observed against predicted ALT on the common site-years for the physics-residual model, Wei et al. and Liu et al., with the RMSE, mean bias and correlation. **c**, Distribution of predicted minus observed ALT on the same site-years (violins clipped at ±80 cm; dot, median; bar, interquartile range). Our models use spatial-block out-of-fold predictions throughout.

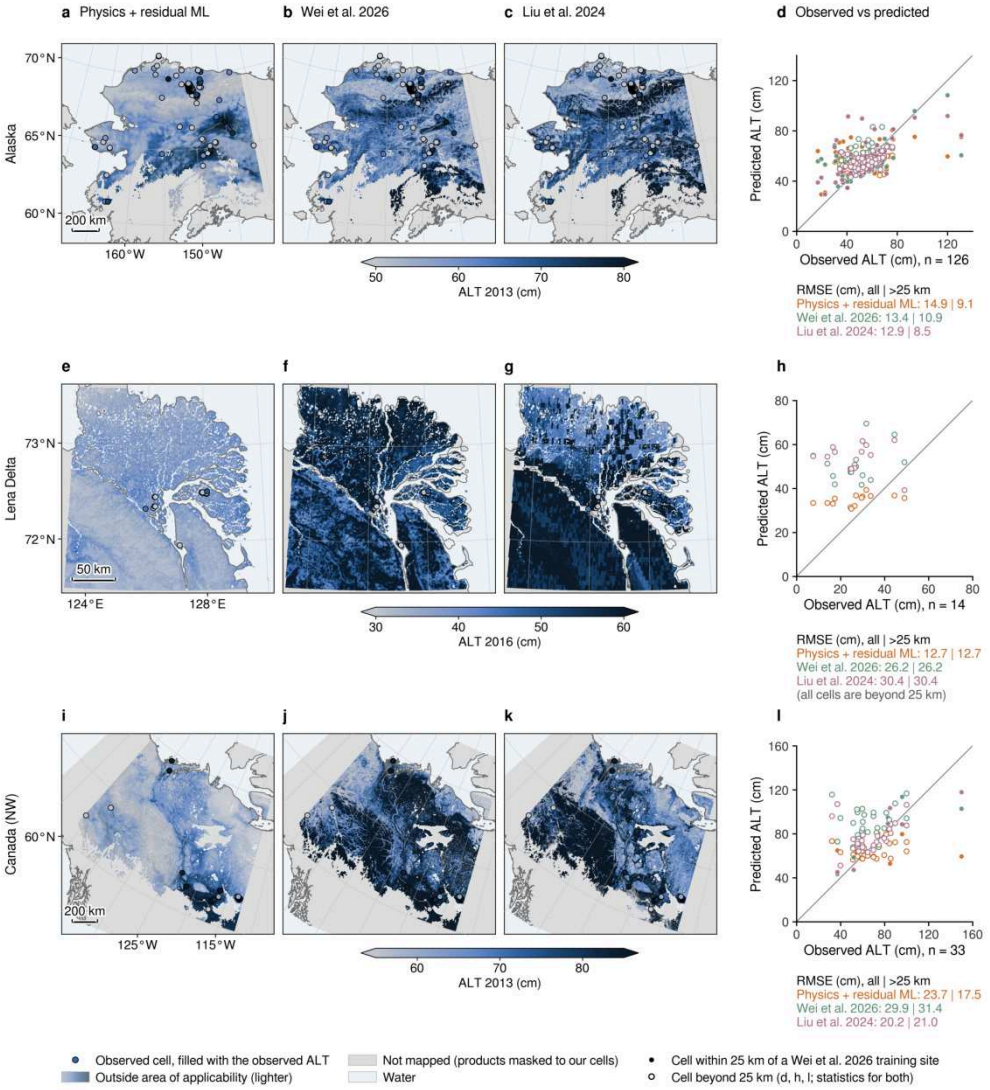


Fig. 6 ALT maps from the physics-residual model and two published products for the year with the most observed cells in each region, with the observed values. a–c, Alaska 2013: physics-residual model, Wei et al. and Liu et al.; the product maps are shown on the cells of our map, and the observed cells are drawn as circles filled with the observed ALT on the same colour scale. **d**, Observed against predicted ALT at the 126 observed cells with values in all three maps (1:1 line); filled markers are cells within 25 km of a training site of Wei et al. and open markers cells beyond, and the RMSE is given below the panel for all cells and for the open cells; our predictions are spatial-block out-of-fold values. **e–h**, The same for the Lena Delta 2016 (14 cells). **i–l**, The same for Canada (NW) 2013 (33 cells). Each region has one colour range anchored on our map (5th–95th percentiles inside the area of applicability, extended to the product medians; product values beyond it are saturated); lighter colours mark cells outside the area of applicability.

Year-labelled 1 km ALT maps, 2011–2024

The physics-residual models fitted to all site-years of each region were applied with the ERA5-Land climate of each year from 2011 to 2024 (Fig. 7) and cover 858,517 cells in Alaska, 38,086 in the Lena Delta and 1,151,169 in Canada (NW). Relative to the 2011–2024 mean, the regional mean anomaly ranged from −2.8 to +6.0 cm in Alaska, from −8.9 to +9.0 cm in the Lena Delta and from −3.8 to +6.4 cm in Canada (Fig. 7d). These anomalies are the response of the model to the climate of each year. The regional mean anomaly correlates with the regional mean TDD to 1 September at 0.92, 0.96 and 1.00 in Alaska, the Lena Delta and Canada (Fig. 7d); the deepest regional thaw occurred in the warmest thaw seasons (2019, 2021 and 2023) and the shallowest in the coldest (2014, 2017 and 2020). In Alaska the anomaly of a warm year is largest in the interior lowlands and smallest on the North Slope (Fig. 7f). Because the set of observed sites changes between campaigns, yearly observed means cannot be compared directly with the mapped anomalies; Supplementary Fig. S9 compares mapped and observed series at long-record sites instead. The linear trend of the yearly maps averaged +2.1, +6.7 and +1.5 cm per decade over the Alaska, Lena Delta and Canada cells, with positive values in 78%, 100% and 91% of the cells (Fig. 7e); it is the modelled response to the ERA5-Land climate of 2011–2024, not an observed trend. The area of applicability covers 56%, 62% and 43% of the Alaska, Lena Delta and Canada cells; cells outside it lie mainly in mountains and at the southern margin of the permafrost zone (Supplementary Fig. S10). The half-width of the 90% prediction interval is 25.5, 24.5 and 41.3 cm, and the dissimilarity d_k of each cell, mapped in Supplementary Fig. S10, indicates the uncertainty of individual cells. Supplementary Fig. S11 shows the residual correction.

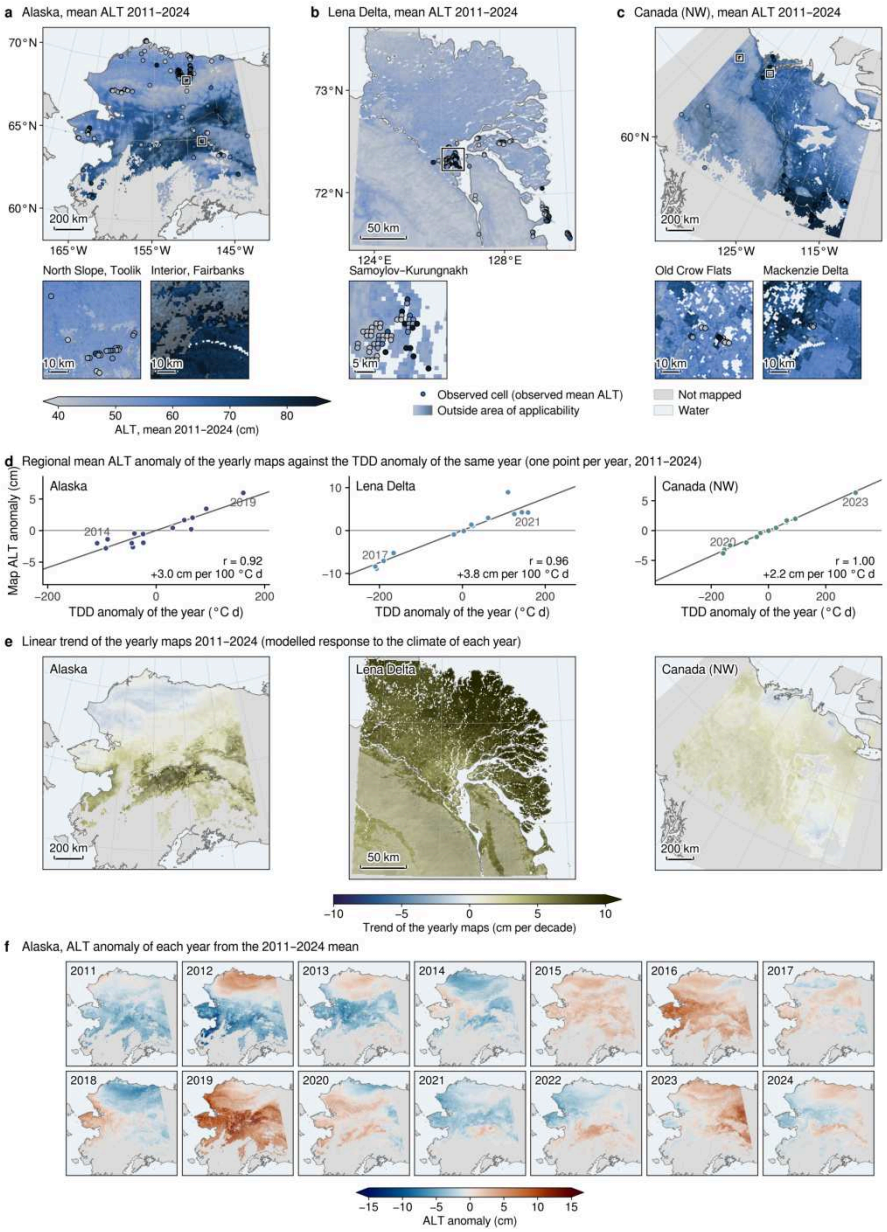


Fig. 7 Year-labelled 1 km ALT maps, 2011–2024. a–c, Mean ALT of 2011–2024 from the yearly maps of Alaska, the Lena Delta and Canada (NW), with windows enlarged at the right (1 km cells) and the observed cells filled with their observed mean ALT; lighter colours mark cells outside the area of applicability. **d**, Regional mean ALT anomaly of the yearly maps against the regional mean TDD anomaly of the same year (one point per year, least-squares line; the warmest and coldest thaw seasons are labelled). **e**, Linear trend of the yearly maps (least squares per cell), which describes the modelled response to the climate of 2011–2024 and is not an observed trend. **f**, ALT anomaly of each year relative to the 2011–2024 mean in Alaska.

Discussion

The main result is that the position at which physical knowledge enters the model determines its accuracy within regions and its transferability between regions. In the physics-residual formulation (equation (5)) the Stefan term represents the response to the thaw-season climate of each year as proportional to the square root of TDD, and the learner corrects the departures from one regional coefficient that soils and terrain impose. This division of labour explains the main patterns of the results. The gain over the Stefan model appeared only when terrain and soil covariates were available, which is consistent with these variables setting the effective coefficient of each site; it increased with the error of the Stefan model; and terrain variables accounted for 50% of the SHAP attribution of the residual model in the Lena Delta and soil variables for 39% in Canada. These patterns indicate that the residual learner captures the part of the thaw depth that one Stefan coefficient leaves unexplained. Direct ML, which must learn the climate response from the labels, had a lower site-year RMSE with thousands of labels, but with 25 labels it was less accurate than the Stefan model in Alaska and Canada and, when the Lena Delta was withheld, its error exceeded that of the Stefan model by 17.8 cm. The in-region RMSE of the physics-residual model exceeded that of direct ML by at most 1.6 cm, while the model kept the transferability of its anchor, which matters where labels are scarce.

In Alaska the residual correction produced no gain, for two reasons. The recalibrated Stefan model already fits the regional level and the year-to-year variation of the dense Alaska labels, so that little systematic residual is left for the covariates to explain, and the error that remains is mostly the spread between probe points inside one 1 km cell (pooled standard deviation 13.0 cm), which no covariate at 1 km resolution can resolve. The absence of a gain in every tercile (Fig. 4b) and the same result on cell-year means (Supplementary Table S5) are consistent with these two reasons and do not indicate a failure of the learner. The learner is therefore not always needed: the spatial-block comparison between the physics-residual model and the recalibrated Stefan model, computed with the labels of a region, shows whether that region has systematic residuals that covariates can explain (a resolved negative ΔRMSE, as in the Lena Delta and Canada) or not (Alaska), and the learner is worth adding only in the first case.

On the same sites and years, the accuracy of the ML products depended on the distance to their training sites, which validations within their own site sets do not assess. Where their training sites are close to our label sites (Alaska, and Canada for Liu et al., whose training data include the earlier release of the same observations), the ML products were as accurate as our model, and the comparison there is not independent. Where they have no nearby training sites (the Lena Delta, median distance 107 km), they predicted between about 40 and 70 cm regardless of the observed ALT, overestimated ALT in the Lena Delta by about 20 cm, most strongly at the thin active layers of the northern delta, and were less accurate than a physics-residual model trained without any Lena Delta label. This pattern matches the error increase of direct ML under region hold-out; we attribute it to products that learn the climate response at training sites elsewhere, although their predictors are yearly. In the physics-residual model the response to the TDD of the stated year is carried by the Stefan term, which

does not depend on the training sites, so this error increase is small. The process-based ESA CCI product, which is not trained on ALT observations, was less accurate than our model in all three regions but, unlike the ML products, was not systematically biased in the Lena Delta (Fig. 5c), which is consistent with its physical basis.

These results extend previous comparisons. Gautam et al.[23] found at 68 Alaska CALM sites of one year that a random forest lost most of its accuracy on held-out sites whereas the Stefan model retained it; this ordering holds here over site-years of 2010–2024 in three regions and under region hold-out, and the two approaches are complementary because a learner that corrects the Stefan model retains its transferability. Studies of physics-guided learning in lakes and rivers reported that process knowledge helps mainly under data scarcity and extrapolation[20–22]; the label-number and region hold-out experiments quantify this for ALT. The finding that random cross-validation underestimates the spatial-block error of direct ML by 3.6–9.9 cm agrees with the literature on spatial validation[10–12]; together with the region hold-out results, this indicates that errors from random or site-based splits do not represent the accuracy of a product in a region without training observations.

The yearly maps apply the physics-residual model to the three regions with an area of applicability and prediction intervals as conditions of use. Each map uses the ERA5-Land climate of one year with the TDD accumulated to 1 September and can therefore be compared with observations or products of the same year. The regional mean anomalies of the maps are the modelled response to the thaw-season climate (correlation with the regional TDD 0.92–1.00). They can be checked against observations only at sites measured year after year (Supplementary Fig. S9). The linear trends of the maps are the modelled response to the climate of 2011–2024, not observed trends. Without local observations, the Stefan model with a coefficient from other regions or the physics-residual model trained on other regions can serve as the starting point; neither showed the large error increase of direct ML in the Lena Delta, although in Canada direct ML trained on other regions was the most accurate. The effect of 25 local labels differed between regions: they reduced the error of the physics-residual model from 23.8 to 19.9 cm in the Lena Delta but increased it from 24.3 to 26.3 cm in Canada, which we attribute to the difference in thaw depth between the two Canadian data sources. The spatial-block comparison within the new region and the area of applicability of the resulting map distinguish these cases.

The study has limitations. Half of the Alaska labels come from 2014, which prevented a resolved estimate of the effect of the year-specific input there, and the Lena Delta labels come from one data set in 16 blocks, which widens the intervals. The labels combine probe transects and CALM grid means from three programmes, and the model error differs between them (Supplementary Table S7), most clearly at the 65 Canadian CALM site-years, whose mean ALT of 89.1 cm is underestimated by −23.0 cm; differences between measurement protocols and site selection are therefore part of the reported errors. The labels are observation sites rather than independent 1 km cells, which the spatial-block design and the cell-year scores account for. The climate input is monthly and the soil covariates are static. The Kudryavtsev model, which includes snow and moisture terms, was more accurate than the recalibrated Stefan model in Canada, and most of the difference between the two anchors there

was corrected by the residual learner; a moisture-aware anchor might further reduce the error that remains after this correction. Active layers thicker than about 3 m lie outside the training range, and the comparison of maps with observations in one year rests on few cells in the Lena Delta and Canada. Because every number is tied to a stated site, year and validation design, the protocol can be applied to new products and regions, and the maps extended year by year.

Conclusions

ALT observations are few and clustered; of the models that extend them to unobserved cells and years, a one-coefficient physical model offers transferability and machine learning offers accuracy near its training sites, and learning the residual of a Stefan model retains most of both. Evaluated on 19,172 site-year observations of three regions with the climate of each observation year, a gradient-boosting model of the residual of a locally recalibrated Stefan model reduced the error of that model by 3.6 cm in the Lena Delta and 3.7 cm in Canada under spatial-block validation. Its gain was resolved in all three regions with 100 local labels, and with 25 labels it was at least as accurate as the Stefan model. With the Lena Delta withheld from training it stayed within 2.0 cm of the Stefan model, whereas the error of direct ML exceeded that of the Stefan model by 17.8 cm. On the same sites and years, the published ML products matched this model only near their own training sites and did not represent the thin active layers of the Lena Delta. The workflow addresses a limitation of each existing approach: it adds to the Stefan model the within-region variation that one coefficient cannot represent, it gives machine learning a physical term that limits the error increase away from the training sites, and its evaluation protocol tests products at stated sites and years outside their training data. The yearly 1 km maps for 2011–2024, with an area of applicability and prediction intervals, provide year-specific thaw depths for ground-stability, hydrological and carbon applications, and in regions with few observations the workflow provides a Stefan-based starting map that improves when local labels represent the region. Daily climate inputs, remotely sensed snow and soil moisture, a moisture-aware anchor and labels from further regions could improve the accuracy and coverage of the maps.

Data availability. All input data are publicly available. ALT observations: ABoVE soil moisture and active-layer thickness, version 2 (ORNL DAAC, https://doi.org/10.3334/ORNLDAAC/2369), ALLena thaw-depth data (PANGAEA, https://doi.org/10.1594/PANGAEA.973813) and GTN-P CALM (PANGAEA, https://doi.org/10.1594/PANGAEA.972777). Climate: ERA5-Land monthly averaged data (Copernicus Climate Data Store). Covariates: Copernicus DEM GLO-30, SoilGrids 2.0, ESA CCI permafrost probability, and MODIS MOD13Q1 and MOD10A2 version 6.1 (accessed through AppEEARS). Products: Wei et al. (https://doi.org/10.5281/zenodo.21667583), Liu et al. (https://doi.org/10.5281/zenodo.10070610), ESA CCI ALT v4 and v5 (CEDA), Aalto et al. (https://zenodo.org/records/5003804) and Yi and Kimball (https://doi.org/10.3334/ORNLDAAC/1760). The site-year label table, the yearly climate-input table, the static covariate table, the out-of-fold predictions

of all methods and the yearly 1 km maps of Alaska, the Lena Delta and northwestern Canada (2011–2024) will be deposited in a public repository upon acceptance [repository and DOI].

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